Action-Faithful Video World Models for Robotics

A focused literature review for an action-counterfactual sensitivity benchmark (Option A). Compiled from Hou et al., *World Model for Robot Learning: A Comprehensive Survey* (arXiv:2605.00080) and the works it cites.

1. Definitions, in build-up order

Each term below uses only the terms defined above it. Read top-to-bottom.

Observation o_t / state x_t. Sensory input at time t (typically RGB image, often multi-view). The *state* is the idealized variable that fully characterizes the system; in practice we use observations as a proxy. Throughout, $x_t \approx o_t$.

Low-level action a_t. Per-step motor command (joint torques, end-effector velocities, gripper open/close). Dense (~10–30 Hz). One value per control timestep.

High-level action / instruction l. Sparse, episode-level task specification, usually a natural-language string. A single l is compatible with infinitely many a-trajectories (e.g. "pick up the red block" admits many grasps and paths).

Trajectory τ. $\tau = (o_0, a_0, o_1, a_1, \ldots, o_T)$. *Real* if produced by a physical / simulated environment; *imagined* if produced by a world model.

Policy $\pi_\theta(a_t \mid o_t, l)$. Parametric map from observation and instruction to motor command. Modern VLA policies internalize l directly.

Inverse Dynamics Model (IDM) $\phi(a_t \mid o_t, o_{t+1})$. Recovers the action that explains a transition. Used by several benchmarks to score whether a generated future is action-realizable.

World model (general). $p_\phi(x_{t+1:t+H} \mid x_t, a_{t:t+H-1}, l)$. A predictive model of agent–environment dynamics. Survey Eq. 1.

Video world model. World model with $x_t = o_t$ in pixel space; output is a future video. Most current robotic WMs are video WMs.

Action-conditioned world model (AC-WM). A WM whose output meaningfully depends on a. The survey calls this the "most important subclass" because action-inconsistent futures are useless for closed-loop control (lines 270–271).

Latent (JEPA-style) world model. Predicts in embedding space rather than pixels: &zcirc;$_{t+1} = f(z_t, a_t)$ with $z_t = E(o_t)$. Cheaper but harder to inspect.

Open-loop rollout. Given o_0 and $a_{0:H-1}$, generate $\hat{o}_{1:H}$ in one shot. No feedback from generated frames into action selection.

Closed-loop rollout. WM is plugged into a control loop; at each step the policy reacts to the WM's prediction. Compounding errors dominate.

Action faithfulness. Property that $\hat{o}_{1:H}$ changes *correctly* when a is perturbed—i.e., the model has internalized $p(\cdot \mid a, \cdot)$ rather than $p(\cdot \mid l, \cdot)$ alone.

Causal conditioning gap. The survey's §8.1 term for the failure mode in which a WM produces intention-consistent (driven by l) but action-inconsistent futures.

Counterfactual sensitivity. Quantitative measure of action faithfulness: how rollout output varies under controlled perturbations of a, with o_0 and l fixed. **This is the variable Option A measures.**

2. Key formulas

Numbered for reference in the paper reviews and gap analysis below.

(F1) WM factorization (Survey Eq. 1): $p(x_{t+1:t+H} \mid x_t, a_{t:t+H-1}, l)$.

(F2) Imagined-rollout joint: $p(\tau) = \pi_\theta(a_0|o_0,l) \prod_t p_\phi(o_{t+1}|o_{\le t}, a_{\le t}, l)\, \pi_\theta(a_{t+1}|o_{t+1},l)$.

(F3) GRPO objective for WM-as-simulator RL (Survey Eq. 18): $L_{RL}(\theta) = -E_t[\, \min(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta),1-\varepsilon,1+\varepsilon)\hat{A}_t)\,]$.

*(F4) **Pixel action sensitivity (proposed):** $S(t; a, a') = E_{o_0,l}[\, \| \hat{o}_t(o_0, a, l) - \hat{o}_t(o_0, a', l) \|_2\,]$. Larger ⇒ more action-faithful when (a′−a) is a meaningful perturbation.*

*(F5) **Counterfactual rank fidelity:** $\rho_{CF} = \mathrm{Spearman}(\, \mathrm{rank}_{sim}\{c(\tau_k^{real})\}, \mathrm{rank}_{WM}\{c(\tau_k^{WM})\}\,)$ over candidate actions $\{a_k\}$ with cost $c(\cdot)$. Generalizes WorldEval to per-action counterfactuals.*

*(F6) **IDM-recoverability** (WorldSimBench, WoW-World-Eval style): $R = E[\, \| a - \phi(\hat{o}_t, \hat{o}_{t+1}) \|_2\,]$.*

(F7) ***a-vs-l reliance gap (proposed):*** $\Delta_a = E\ ||\hat{o}(a,l) - \hat{o}(a',l)||$ *(random a');* $\Delta_l = E\ ||\hat{o}(a,l) - \hat{o}(a,l')||$ *(paraphrased l').* ***Action-Reliance Ratio*** $ARR = \Delta_a/\Delta_l$*; ARR>1 ⇒ action-faithful, ARR<1 ⇒ instruction-driven.*

3. Paper reviews (only what matters for Option A)

3.1 Foundations — concept anchors

Ha & Schmidhuber (2018), "Recurrent World Models Facilitate Policy Evolution." Origin of the modern term. Trains a recurrent latent dynamics model (V–M–C: vision, memory, controller) and shows that policies trained *inside imagined rollouts* of the learned dynamics transfer to the real environment. **Take-away:** the legitimacy of evaluating policies in imagined rollouts depends entirely on whether the WM tracks action consequences — the unstated assumption your benchmark interrogates.

Hafner et al., DreamerV3 (2023). Canonical model-based RL with a Recurrent State-Space Model (RSSM); shows scaling/robustness across hundreds of tasks. **Take-away:** establishes the latent-state baseline; recent video WMs are pixel-space generalizations, but they have not inherited Dreamer's careful action-conditioning evaluation.

Yang et al., UniSim (ICLR 2024). First widely-cited paper to frame video generation as an interactive simulator: p(video$_{t+1}$ | video$_t$, action). **Take-away:** defines the genre your benchmark evaluates, but offers only appearance- and task-success-style metrics.

Assran et al., V-JEPA 2 (2025) & Mur-Labadia et al., V-JEPA 2.1 (2026). Latent self-supervised video models; V-JEPA 2.1 shows zero-shot robot planning with image goals via latent action-conditioned rollout. **Take-away:** a strong counter-argument to pixel WMs. If your benchmark shows pixel WMs are not action-faithful, JEPA-style models become the natural alternative; cite as the contrast class.

3.2 Action-controllable video world models — your probe subjects

IRASim (Zhu et al. 2025). Trajectory-to-video formulation. Innovation: *frame-level action conditioning* — each transformer block injects the action of *that* frame rather than a single global condition. Trained on Bridge / RT-1; code released. **Strength:** the cleanest published architecture for action-faithful video. **Limitation:** evaluation uses FVD + a small in-paper visual comparison, not a controlled counterfactual probe. **Probe priority: HIGH.**

Ctrl-World (Guo et al. 2026, ICLR). Designed *for policy evaluation*: multi-view prediction, frame-level action control, memory module for long-horizon consistency. Shows that action-faithful rollouts can rank policies. **Strength:** author claims action faithfulness *and* measures policy-rank preservation — the closest published precedent for what you want to do. **Limitation:** author-defined metrics, single benchmark, no perturbation suite. **Probe priority: HIGH; primary baseline.**

Vid2World (Huang et al. 2026, ICLR). Recipe for converting any pretrained video diffusion model into an interactive WM via causal masking + action-conditioning layers. **Strength:** plug-and-play, leverages public video-diffusion checkpoints. **Limitation:** conversion does not *verify* that the resulting model is action-sensitive. **Probe priority: MEDIUM.**

EnerVerse-AC (Jiang et al. 2025). Multi-view action-conditional generator that doubles as a data engine + evaluator. **Strength:** evaluation utility is a first-class objective. **Limitation:** reuses task-success metrics. **Probe priority: MEDIUM.**

Cosmos Predict 2.5 / Cosmos Policy (NVIDIA, Ali et al. 2025; Kim et al. 2026). Foundation-scale (multi-billion-parameter) video WM with public checkpoints; Cosmos Policy fine-tunes for visuomotor control. **Strength:** size + availability. **Limitation:** trained primarily from text/task signal — the prime suspect for high-l, low-a behavior. **Probe priority: HIGH for foundation-scale comparison.**

Genie Envisioner (Liao et al. 2026). Unified world-foundation platform integrating video WM + action decoding. **Strength:** conceptual peer of Cosmos. **Limitation:** same training-signal bias. **Probe priority: LOW** (use as confirmation, not primary subject).

3.3 Existing evaluation benchmarks — your prior art

WorldEval (Li et al. 2025). Asks whether a WM preserves the relative ranking of policies in real vs. imagined rollouts. **Strength:** directly measures decision-utility. **Limitation:** confounds policy quality with WM quality — a WM that ignores a entirely could still preserve ranking if all policies in the test set are similar in dynamics. No isolated action probe.

WorldGym (Quevedo et al. 2025). Treats WM as a Monte-Carlo evaluation environment; checks whether estimated policy *value* matches real value. **Limitation:** aggregate value over an episode; no per-step or per-action diagnostic, no controlled perturbation.

WorldSimBench (Qin et al. 2025). Two-track: (i) perceptual fidelity, (ii) *manipulative evaluation* via inverse-dynamics recovery (F6) — given $\hat{o}_{t:t+1}$, run an IDM, check whether recovered actions match input. **Closest existing precedent** for action faithfulness. **Limitation:** uses a single fixed IDM; tests whether *some* action could explain the rollout, not whether the rollout responded to *this specific* action.

WorldArena (Shang et al. 2026). Multi-role evaluation: synthetic data generation, policy evaluation, action planning. **Limitation:** each role scored separately; no isolated causal-conditioning probe.

World-in-World (Zhang et al. 2025). Closed-loop benchmark; reports end-task success when WM is plugged into a planner. **Limitation:** compounded failures invisible — if the WM is action-blind but the policy is robust, you cannot tell.

WoW-World-Eval / "Wow, wo, val" (Fan et al. 2026). Comprehensive embodied WM Turing-style test, includes IDM-based executability. **Closest competitor. Limitation:** holistic Turing judgment; no factor-resolved sensitivity curve and no controlled action-perturbation suite.

WM-ABench (Gao et al. 2025, "Do VLMs have internal world models?"). Decomposes evaluation into *atomic* capabilities: spatial, temporal, mechanistic, counterfactual. **Methodologically the closest** to what you want, but applied to **VLMs, not video WMs.** Transferring the atomic-capability framework to video WMs is a publishable framing on its own.

EWMBench, RBench, DreamGen-Bench, EVA-Bench. Open-loop predictive benchmarks with embodied flavor (scene/motion/semantic factors; structure+physics+action-completeness; instruction following; long-horizon OOD). **Common limitation:** ultimately appearance-driven; action sensitivity not isolated as a primary axis.

3.4 Critique allies — your motivation

WorldVLA (Cen et al. 2025). Unified VLA + WM. The authors *explicitly diagnose* "weak action conditioning" as the bottleneck and add a unified-training fix on the *training* side. Your contribution complements theirs on the *evaluation* side.

WoVR (Jiang et al. 2026). Treats WM reliability as the central bottleneck in WM-as-simulator RL. Key claim: hallucination "does not merely reduce visual quality, but can directly corrupt the assessment signal itself" (survey lines 1027–1029). Quote-worthy.

EVA (Wang et al. 2026). Aligns video WMs with executable robot actions via inverse-dynamics rewards. Frames the issue as the "executability gap."

Causal WM (Li et al. 2026, arXiv:2601.21998). Explicitly causal framing of WM training. Cite for terminology.

Pan et al. 2026, "Much ado about noising." Skeptical position paper challenging diffusion-based generative robot control hype. Cite as the balancing voice; demonstrates you are aware of the controversy.

4. Gap matrix

Compact strengths / limitations / pointed-to extensions for every paper above. Patterns repeat — that's the point of the synthesis below.

Paper	Role	Strength	Key limitation	Implied extension
Ha & Schmidhuber 18	Found.	Imagined rollouts can train policies	Latent only; toy domains; no faithfulness audit	Pixel-space + action-faithfulness audit
DreamerV3	Found.	Stable model-based RL with RSSM	Small latent state; no video extension	Video-scale faithfulness benchmark
UniSim	Found.	Frames video gen as interactive simulator	FVD / task-success metrics only	Counterfactual action probes
V-JEPA 2.1	Found.	Latent action-cond. zero-shot planning	No head-to-head with pixel WMs	Cross-modality faithfulness comparison
IRASim	AC-WM	Frame-level action conditioning	Single-view, single benchmark	Standardized counterfactual eval
Ctrl-World	AC-WM	Multi-view + memory + author rank-test	Author-defined metric; no perturb suite	External standardized benchmark
Vid2World	AC-WM	Recipe to convert any video diffusion WM	Conversion not verified action-sensitive	Post-conversion sensitivity audit
EnerVerse-AC	AC-WM	Eval-aware multi-view design	Reuses task-success metrics	Atomic action-cond. metrics
Cosmos Predict / Policy	AC-WM	Foundation scale, public checkpoints	Trained mostly from text/task signal	*a*-vs-*I* disentanglement test
Genie Envisioner	AC-WM	Unified WM+action platform	Same training-signal bias	Same a-vs-I probe
WorldEval	Eval	Real-vs-imagined policy ranking	Confounds policy and WM quality	Decoupled WM-only causal probe
WorldGym	Eval	MC value-fidelity test	Aggregate; no per-step diagnostic	Per-action counterfactual

Paper	Role	Strength	Key limitation	Implied extension
WorldSimBench	Eval	IDM-recovery test (closest precedent)	Single fixed IDM; not a perturbation test	Multi-IDM ensemble + controlled perturb.
WorldArena	Eval	Multi-role functional evaluation	No isolated causal probe	Cross-role causal diagnostic
World-in-World	Eval	Closed-loop end-task success	Compound failures hide root cause	Decomposed mid-loop diagnostic
WoW-World-Eval	Eval	Holistic IDM Turing test	Not factor-resolved over horizon	Sensitivity curve S(t) over time
WM-ABench	Eval (VLM)	Atomic capability decomposition	VLMs not video WMs	Port atomic decomposition to video WMs
EWMBench / RBench / DG-Bench / EVA-Bench	Eval (open-loop)	Embodied open-loop predictive scoring	Still appearance-driven aggregates	Action-isolated factor in the score
WorldVLA	Critic	Diagnoses weak action-conditioning	Fix is in training, not in eval	Eval-side benchmark of weak a-cond
WoVR	Critic	Reliability is central bottleneck	Reliability not directly measured	Direct reliability probe
EVA	Critic	Frames executability gap explicitly	Aligns via training only	Eval-side executability metric
Causal WM (Li 26b)	Critic	Causal framing of WM training	No empirical eval of non-causal WMs	Empirical causal-deficiency benchmark
Pan et al. 26	Critic	Skeptic position on diffusion control	Not a benchmark	Quantitative test of skeptic claims

5. Cross-paper patterns (the recurring story)

Pattern 1 (across all six AC-WMs). Every action-controllable video WM *asserts* action-conditioning is essential and adds an architectural mechanism for it (frame-level injection, causal mask, multi-view, memory). **None verifies the mechanism worked with a controlled action counterfactual.** Verification is left to FVD-like appearance scores or downstream task success.

Pattern 2 (across all eight evaluation benchmarks). Every benchmark measures an *aggregate* quantity — success rate, value, rank correlation, FVD, IDM-recovery error — computed over whole episodes or whole datasets. **None holds (o_0, l) fixed and perturbs a in a controlled way.** None reports sensitivity as a function of horizon t. None uses an ensemble of IDMs to robustify the recovery test against IDM idiosyncrasy.

Pattern 3 (across all five critic papers). Everyone agrees weak action conditioning is the central bottleneck. The fixes are uniformly on the **training side** (unified objectives, IDM-reward alignment, causal losses). **No one provides a standardized, model-agnostic, training-free benchmark for measuring how weak the conditioning is on a given checkpoint.**

Pattern 4 (across all foundations). The legitimacy of every world-model use case — imagination-based RL, policy evaluation, plan ranking — rests on the assumption that the WM tracks action consequences. **That assumption has never been directly stress-tested at the level of individual checkpoints.**

The gap, in one sentence. Across foundations, AC-WMs, eval benchmarks, and critics, the missing artifact is a *training-free, controlled, horizon-resolved counterfactual probe of action sensitivity* that any released checkpoint can be scored on in an afternoon. The community needs this; nobody has built it.

6. Two specific, implementable gaps for Option A

Gap 1 — Action-Counterfactual Sensitivity Benchmark (ACSB)

What it measures. Hold (o_0, l) fixed; for each scripted scene with ground-truth action a, run four perturbations: (a) *reverse*, (b) *zero*, (c) *random Gaussian* matched to a's marginal, (d) *swapped* — the action sequence from a different scene with the same instruction. Roll out each in the WM and the simulator. Report:

• **S(t)** from F4 as a function of horizon t. Curve, not scalar.

• **Pearson(WM-divergence, sim-divergence)** across perturbations: does the WM diverge in the directions and magnitudes the simulator does?

• **Counterfactual rank fidelity** ρ_{CF} from F5 across the four variants.

Why this is the right gap. It directly answers the question every paper in the gap matrix dances around: *does this checkpoint actually use a?* It is training-free (inference only), model-agnostic (works on any WM with an action input), and produces a single standardized score that can be reported alongside FVD.

Concrete scope (6 weeks, one A100). Scenes: 30 from LIBERO-Long (language-conditioned, long horizon, ground-truth simulator). Subjects: Ctrl-World, IRASim, Cosmos Policy. Perturbations: 4 above. Seeds: 5. Total rollouts: 30 × 3 × 4 × 5 = **1,800**. Easily fits.

Strongest version of the result. If foundation-scale WMs (Cosmos) show low S(t) for t>1s while smaller dedicated AC-WMs (IRASim, Ctrl-World) show high S(t), you have a publishable finding: *scaling the video backbone does not buy you action faithfulness*. That headline alone clears the bar for the ICLR World Models workshop or CoRL 2026 workshop.

Gap 2 — Action-Reliance Ratio (ARR), F7

What it measures. Hold one input fixed and perturb the other, with the natural perturbation in each case: random a′ for a; LLM-paraphrased l′ for l. ARR = Δ_a/Δ_l. **ARR>1** means the model uses a more than l (action-faithful); **ARR<1** means it uses l more than a (instruction-driven; the survey's §8.1 failure mode).

Why this complements ACSB. ACSB asks *how much* the model responds to a. ARR asks *which signal it relies on more*. Together they distinguish "action-blind" from "instruction-driven": a model can be responsive to action perturbations *and* still be more responsive to instruction paraphrases — that's a subtle failure ACSB alone misses.

Concrete scope. Same scenes/subjects as ACSB. Add 5 LLM paraphrases per instruction (Claude or GPT-4 API; trivial). Same total rollouts.

Recommended order

Build ACSB first (Gap 1). It has a cleaner story (one axis of variation, one headline number). Reserve ARR (Gap 2) for the extended-paper version. Pitch to ICLR 2027 World Models Workshop (CFP usually opens November/December); CoRL 2026 robot-learning workshops are the fallback. Both accept 4–8-page submissions, which is the right size for a single benchmark contribution.